INCH-POUND

MIL-P-53108(ME)
26 October 1990
SUPERSEDING
(see 6.7)

MILITARY SPECIFICATION

PUMP UNIT, CENTRIFUGAL, DIESEL-ENGINE-DRIVEN,

800 GPM AT 1800 FEET TOTAL HEAD

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid-mounted, diesel-engine-driven, multistage, centrifugal pump.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-W-572	- Wood Preservative: Water-Repellent.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronics and Associated Equipment).
MIL-H-370	- Hoses and Hose Assemblies, Nonmetallic: Rubber, Liquid Fuel.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-S-2940	- Snubbers, Fluid Pressure, Instrument Protection.
MIL-G-3859	- Grease Guns, Hand Operated, Lever, Push and Screw Type, General Specification for.
MIL-E-10062	- Engine: Preparation for Shipment and Storage of.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-14105	- Paint, Heat-Resisting (for Steel Surfaces).
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-S-45180	- Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-C-62122	- Cable Assembly, Intervehicle Power; Plug and Receptacle.
MIL-T-83133	- Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35.
MIL-B-83769	- Batteries, Storage, Lead-acid, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1410	-	Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel.
MIL-STD-1791	-	Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS24532	-	Voltmeter: Battery-Generator, 28 Volt DC.
MS24540	-	Indicator, Pressure, 24 Volt, DC, Range 0-120 PSI.
MS24543	-	Indicator, Temperature, Electrical Resistance: 24 Volt DC, Range 120-240 °F.
MS27021	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Hose Shank, Type II.
MS35643	-	Gasket: Synthetic Rubber, Cap, Fuel Tank.
MS35644	-	Filler Neck: Fuel Tank-Military Vehicles.
MS35645	-	Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35829	-	Hinge, Butt, Continuous (Piano) .075 Thickness, .187 and .250 Pin Dia.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

MILITARY TRAFFIC MANAGEMENT COMMAND (MTMC/TEA)

Pamphlet 70-1 – Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5	-	Pipe Flanges and Flanged Fittings.
B40.1	-	Gauges-Pressure Indicating Dial Type-Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

AMERICAN PETROLEUM INSTITUTE (API)

Std 610 - Centrifugal Pumps for General Refinery Service.

(Application for copies should be addressed to the American Petroleum Institute, 1220 L Street, Northwest, Washington, DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pump unit shall be a skid-mounted, diesel-engine-driven, horizontal split case, multistage, centrifugal pump (see figure 1 and 6.7). The unit shall have an operational point of 800 gpm at 1800 feet total head when pumping diesel fuel with a specific gravity of 0.85 (7.08 lb/gal).

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. The material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

3.3.1 Material deterioration prevention and control. The pump unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the unit may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with each component and subcomponent, and shall make this information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pump unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the unit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pump unit.

3.4 Design. The pump unit shall be designed and constructed in accordance with all applicable requirements and provisions of API Std 610 except for 3,000 hours of uninterrupted operation.

3.5 Interchangeability. All pump units furnished with similar options under a specific contract shall be identical, to the extent necessary, to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Operating conditions.

3.6.1 Operating temperature. The pump unit shall perform as specified herein in any ambient temperature from -25 to +135 °F.

3.6.2 Altitude. The pump unit shall perform at any altitude from sea level to 6,000 feet above sea level at 100 °F, without significant loss (not to exceed 10 percent) of capacity and head.

3.6.3 Storage temperature. The pump unit shall not be damaged by storage in any ambient temperature from -60 to +155 °F.

3.7 Safety. All shafts or rotating parts of the pump unit shall be guarded when such parts are exposed to contact with personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact with personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Adequate shielding may be used to attain the same result. Exhausts or discharges from the equipment shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

be directed so that they do not endanger personnel. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are necessary and are not detrimental to safety.

3.8 Human factors engineering. The pump unit shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given to general requirements, control/display integration, visual displays, controls, labeling, anthropometry, design for maintainability, and hazards and safety as specified in MIL-STD-1472 4.1 thru 4.9, 5.1, 5.2, 5.4, 5.5, 5.6, 5.9, and 5.13.

3.9 Noise limits. Noise levels produced by the pump shall comply with the requirements of MIL-STD-1474, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, and except as specified herein.

a. The noise level at the operator's position, defined to be 18 inches from the control panel, and other personnel occupied areas of the pump, defined to be 1 meter from any point on the perimeter of the pump at locations other than the operator's position, shall not exceed category D of table I [85 dB(A)]. Where the limit of category D can be documented as being clearly beyond the state-of-the-art (see 3.9.1.2), selection of another noise limit shall be made by the procuring activity and the above requirements shall again apply.
b. The following table shall be used in lieu of MIL-STD-1474, table II.

TABLE I. Steady-state noise limits [dB(A)] for personnel-occupied areas.

	Limit Category					
	A	B	C	D	E	F
A-weighted Limit [dB(A)]	107	100	90	85	75	65

3.9.1 Acceptance.

3.9.1.1 Compliance. Compliance with the required noise limits shall be documented based upon test requirements of 4.6.2.8 and the data recording requirements of MIL-STD-1474, 5.5.

3.9.1.2 Non-compliance. Non-compliance with the required noise limits shall be documented based upon test requirements of 4.6.2.8 and the data recording requirements of MIL-STD-1474, 5.5. Additionally, the following shall apply:

a. The required maximum noise limit(s) may be established to be technically infeasible based upon testing and analysis which clearly and convincingly proves that achievement of the required limit(s) is beyond the state-of-the-art for the item/system being procured. Such testing and the resultant analysis shall be conducted and reported by a professionally qualified acoustical consultant or acoustical laboratory. Documentation shall be furnished to the procuring activity for consideration of whether or not increase of the required limit(s) is justified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

b. Documentation shall contain technically defensible data including technically supported design considerations, technically supported design recommendations for noise reduction, and technically supported predictions of the resultant noise levels.
c. Noise reduction feasibility documentation shall also include the following:
    (1) Identification of all the noise sources that contribute to the noise level at the locations of interest.
    (2) Identification of all noise paths between the noise sources and the locations of interest.
    (3) Rank ordering of the source/path combinations in terms of their contribution to the overall level at the locations of interest.
    (4) Development of noise control measures for each source/path combination in order of dominance until the required noise levels at the locations of interest are attained.
    (5) Clearly stated, convincing, and technically/fiscally supported trade-off analyses of noise control benefit against other design requirements such as weight, access, etc.
    (6) An experimental procedure known as "window analysis" shall be used in steps (1) through (5). The procedure involves measuring the noise level while eliminating all noise sources and paths except the single one of interest. For example, one type of engine noise source determination could be made by running the engine under load with and without the engine fan being driven.
    (7) The window analysis procedure shall be repeated for each of the combinations [see (1) to (3) above] identified.
d. If, and only if, the maximum noise limit(s) required has been convincingly documented as being clearly beyond the state-of-the-art, may the procuring activity expressly grant the contractor written permission to exceed the specified limit(s).

3.10 Performance. The pump unit shall be operable at various speeds to deliver product flow and feet of head that includes a design point of 800 gpm at 1800 feet total head (0.85 specific gravity). The pump unit shall deliver all required capacities, including the design point capacities, when any axis of the pump unit is positioned at an angle of five degrees from the horizontal.

3.11 Construction. The pump shall comply with American Petroleum Institute (API) Std 610 for the applications and operating conditions specified in this specification.

3.11.1 Construction materials. The pump shall be constructed from the materials given in API Std 610, table H-1, material class S-6. All chrome parts shall be hardfaced.

3.11.2 Seals and bearings. The pump shall be equipped with balanced mechanical seals and antifriction bearings for durability and reliability.

3.11.3 Impellers. The pump shall be equipped with opposed impellers to balance hydraulic thrust. The impellers shall be positioned on a step-locked shaft and secured against axial movement in either direction along the shaft. Impellers shall be dynamically balanced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

3.11.4 Suction and discharge connections. Suction and discharge connections shall be 6-inch 300 pound raised face flanges, in accordance with ANSI B16.5, and shall be fitted with steel blind flanges including suitable gaskets to protect the pump connections during shipment and periods of non-use. The pumps shall be furnished with pipe adapters for the suction and discharge connections having a 300 pound raised face flange connection on one end and an Inland Petroleum Distribution System (IPDS) 6 inch single groove connection, as shown in figure 2, on the other end. The adapters shall be designed so that when attached to the suction and discharge connections, the IPDS single groove ends are positioned 9.5 $\pm$ 1.5 inches vertically from the bottom of the skid. If the suction adapter is not identical to the discharge adapter, each shall be labeled accordingly.

3.11.5 Casing vent lines. The pump casing vent lines shall be routed to a point at the edge of the skid base and shall terminate in a single 3/4-inch pipe. The height of the pipe shall be sufficient to allow a 10-inch vessel to be placed underneath.

3.11.6 Tapered pins. A tapered pin shall be provided on each of the two diagonally opposite pump mounting feet to establish a bench mark for pump-driver alignment.

3.12 Engine. The engine shall be an industrial, turbocharged, diesel engine, selected in accordance with MIL-STD-1410, class II. The engine shall deliver power at its continuous duty rating to meet the pump requirements of 3.10. The engine shall be capable of operating on all fuels specified in VV-F-800 and with JP-8 conforming to MIL-STD-83133 at all ambient temperatures specified in 3.6.1 without degradation of reliability or durability. The engine shall be equipped with, but not limited to, the following:

a. A 24-volt electric cranking system with a charging alternator and batteries suitable for -25 °F starting.
b. Heavy duty air cleaner, dry type, replaceable cartridge, with restriction indicator.
c. Fuel filters, primary, water separating type; secondary, with drain cock.
d. Manual engine controls to include a waterproof pushbutton starter switch and a push-pull shutdown control (pull to stop).
e. Automatic engine shutdown controls in the event of engine overspeed, high coolant temperature, low lubricant oil pressure, or pump low suction pressure (device with an adjustable working range of 0-50 psig.
f. Spark-arresting muffler with guard.
g. Drain tube to permit draining the engine oil into a container without soiling other components of the system.
h. Auxiliary charging receptacle conforming to MIL-C-62122 for recharging the batteries from a remote power source without operation of the engine.
i. Cold weather starting aid conforming to MIL-E-52649, type 3.
j. Port to sample engine oil for testing.
k. Provisions to lubricate the turbocharger bearing prior to startup. Lubrication of the bearing shall be accomplished without tools or any degree of disassembly and shall utilize oil already in the engine lubricating oil system. The method of bearing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

lubrication shall not require energizing the engine starter motor. Engine oil to be at operational level; this control shall be mounted in the instrument panel.

l. Heavy duty cooling system, shall be capable of maintaining the manufacturer's recommended coolant temperature while operating at rated capacity and head in an ambient temperature of 120 °F. The radiator shall have provisions for cleaning from the front, using air.
m. Instrumentation consisting of lubricating oil pressure gage, coolant temperature gage, tachometer-hourmeter, battery charging indicator and an air cleaner restriction indicator.
n. Engine mounting equipped with a minimum of 4 axial and 4 lateral jacking screws to facilitate alignment of the pump and engine under field conditions.

3.12.1 Engine enclosure. The engine shall be provided with a sheet metal, noise abating enclosure that shall enclose the engine sides, top (except for the exhaust connections), and ends (except for openings for cooling air). The enclosure shall be designed for maximum accessibility for ease of maintenance or replacement of components. Doors or access panels shall be provided as necessary and shall be fitted with flush-type latches, clamps, or hinges and shall be self-supporting in the open position. Spring-loaded side panels shall not be used. The enclosure shall be utilized to the maximum extent in control of temperatures around pump unit components and in noise control. Ambient temperatures around engine accessories, batteries, and electrical components shall not exceed 160 °F for any load in an ambient temperature of 120 °F. Covers shall be provided for air intake and exhaust if necessary. The pump shall be mounted externally to the enclosure. All doors and access panels which should be closed or in place when the pump unit is operating shall be conspicuously marked to indicate this requirement as specified in 3.8.

3.12.2 Connect/disconnect clutch. The engine shall be equipped with a connect/disconnect clutch that will permit operation of the engine without the pump operating to allow for engine warmup.

3.13 Automatic engine/pump controller. The pump unit shall be equipped with an automatic engine/pump controller. Throttle control for the pump unit shall be provided exclusively by the automatic engine/pump controller. The automatic engine/pump controller shall provide for two modes of operation: (1) Manual engine speed control, (2) Discharge and suction pressure control.

3.13.1 Operator controls. The operator controls shall be located on the instrument panel, similar to the one shown in figure 3, and shall consist of an adjustable two-position selector switch and a set point adjustment. The operator's controls shall be labeled as to function as specified in 3.8. The adjustable two-position selector switch shall provide operation in either of the two modes "manual engine speed control" or "automatic discharge pressure control". Engine starting shall be possible only when the two-position switch is in the manual engine speed control position. The second operator control shall be an adjustable device that will function as a set point adjustment. When the two-position switch is in the manual engine speed control position, the set point adjustment shall establish the operating speed of the engine selected by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

operator. When the two-position switch is in the automatic discharge pressure control position, the set point adjustment will establish the discharge pressure delivered by the pump as selected by the operator. Rotation of the set point adjustment shall be clockwise going from lowest to highest setting.

3.13.2 Manual engine speed control. In the manual engine speed control mode position, the pump speed shall be isochronously controlled; that is, the engine/pump speed shall stay at the speed selected by operating personnel using the "set point adjustment", irrespective of pump discharge pressure, up to the maximum allowable engine speed provided none of the other shutdown devices (high coolant temperature, low lubricant oil pressure, or low pump suction pressure) have operated. The "manual engine speed control" mode position must be used to permit engine starting and the low range of the "set point adjustment" device used for engine warm up.

3.13.3 Automatic discharge pressure control. In the "automatic discharge pressure control" mode, the automatic engine/pump controller shall operate in conjunction with an electronic speed control governor, which would regulate engine speed to maintain the respective discharge pressure corresponding to "set point adjustment" position selected by the operator. The suction pressure control function shall override the discharge pressure control function in the event adequate suction pressure is not available to the pump to achieve the desired discharge pressure. The suction pressure set point adjustment shall be accessible from the side or back of the controller for adjustment by personnel using common hand tools.

3.13.4 Pressure ranges. The discharge pressure controller and associated transducers shall have an adjustable working range of 0-900 psig. The suction pressure control and associated transducer shall have an adjustable working range of 0-150 psig. The automatic engine/pump controller shall be factory set to the following values: low pressure override – 20 psig; maximum discharge pressure – 760 psig.

3.14 Instrument panel. The pump unit shall incorporate a steel instrument panel, rigidly mounted on the suction side of the pump in a readily accessible area to enable the operator to read the pump gages, perform operational procedures and the required maintenance to the controls and indicators, and to repair or replace the panel if required. The panel shall be completely enclosed with a hinged door with locks and shall be weatherproof. The panel shall include sufficient lighting to allow the operator to read all pump and engine gages and operate all pump and engine controls in otherwise total darkness. Replacement of light bulbs shall require no tools. Instrumentation shall include, but not be limited to, an electric tachometer/hourmeter, a voltmeter gage conforming to MS24532-2, a lube oil pressure gage conforming to MS24540-2, a coolant water temperature indicator gage conforming to MS24543-2, two shock-, vibration- and corrosion-resistant pressure gages conforming to ANSI B40.1. The suction gage shall have a pressure range of 0-400 psi and the discharge gage shall have a range of 0-1000 psi. The suction gage shall have an overpressure range of at least 900 psi and the discharge gage shall have an overpressure range of at least 1500 psi. The dials shall be white with black calibration lines and figures and shall be not less than 4-1/2 inches in diameter. All gages shall be labeled according to function (i.e., TEMPERATURE, ENGINE SPEED, etc.) and color coded as specified in 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

The pressure gages shall be equipped with fluid dampeners conforming to MIL-S-2940, 2-L-a-P-G. All gage lines shall contain shutoff valves and be located and protected to prevent damage by personnel while handling, transporting, operating, and servicing. Pressure gage lines shall have provisions for complete drainage to prevent freezing and shall provide for air elimination during startup. The air eliminator shall not require tools to be operated. The system shall be equipped with shut down lights for each required emergency shutdown. The lights shall incorporate a manual reset latch for the low suction pressure and overspeed fault circuits. The low pressure shutdown circuit shall incorporate a time delay (adjustable from 0.1 to 30 seconds) that allows for small variations in suction pressure.

3.15 Fuel system. The steel fuel tank shall have sufficient capacity to provide 4-hour operation without refueling. Two 3-way fuel selector valves with corrosion-resisting faceplates with markings of “OFF”, “UNIT TANK”, and “AUXILIARY” and two 10-foot long fuel hoses conforming to MIL-H-370 shall be furnished to permit engine operation from the fuel tank or an auxiliary fuel service. One end of each hose shall be equipped with a coupling conforming to MS27021. The valve shall be clearly marked to indicate its operating position. The valve control shall be provided with a double-ended arrow showing direction of operation and labeled at each end to indicate the functional result (i.e., open, close, etc.). The fuel tank shall be equipped with a mechanically actuated fuel gage graduated from empty to full, a filler neck conforming to MS35644, a tank cap conforming to MS35645, and a gasket conforming to MS35643. The label “DIESEL OR JP-8 ONLY” shall be placed on or near the fuel fill port as specified in 3.8.

3.16 Skid mount. The pump unit shall be securely mounted to a heavy duty structural steel skid assembly. The skid mount shall consist of two steel skid runners with a heavy duty baseplate for mounting components of the pump unit. The runners shall incorporate enclosed forklift pockets on each side of the base and shall be upturned or beveled at each end to aid in skidding the unit into position. The forklift pockets shall have dimensions of 6 inches by 20 inches. The skid base shall allow for four-way entry by a forklift. The skid runners shall be furnished with mounting holes for anchoring the units to a concrete foundation or wood deck. The number and size of holes required shall be as determined by the manufacturer for anchoring the unit under the most severe conditions of vibration and thrust likely to develop during operation. The skid runners shall be of sufficient height to provide ready access to any connections located under the units and shall be of sufficient width to provide adequate bearing surface on a smooth, gravel base. The base assembly shall incorporate provisions to mount the ground rods, battery box assembly, and the storage box assembly.

3.17 Transportability. The pump unit shall be equipped with tiedown and slinging provisions as specified herein. The pump unit(s) shall be capable of being transported by highway, rail or air as specified herein.

3.17.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.17.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class l or 3, type IV. The provisions shall enable the complete pump unit(s) to be lifted in the normal operating position. The provisions shall be located so that not less than 1-inch (2.54 cm) clearance is

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.17.1. All slinging/tiedown provisions shall be labeled "LIFT HERE", "TIEDOWN HERE" or "LIFT/TIEDOWN HERE", as appropriate, in 1-inch (2.54 cm) high letters.

3.17.3 Air transportability. The pump unit(s) shall meet the requirements of MIL-STD-1791 for air transport. When required (see 6.2), the contractor shall provide sufficient data to the Government to confirm that the pump unit being supplied is capable of being loaded and transported in the C-130, C-141, and C-5A aircraft and lifted by CH-47 helicopters both internally and externally (see 6.6.1).

3.17.4 Rail transportability. The pump unit(s) shall be rail transportable in CONUS and NATO countries without restrictions. The pump unit(s) shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MTMC/TEA pamphlet 70-1 outline diagram when loaded on a 50-inch (127 cm) high rail car. The pump unit(s) shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.6.2.10.4 without failure, damage, or permanent deformation.

3.17.5 Highway transportability. The pump unit(s), when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States of the United States of America.

3.18 Grounding assembly. Two 3/4-inch nominal diameter ground rods, 60-inches long, of copper clad, cold-rolled steel shall be provided. Two uninsulated, braided ground wires, 10 American Wire Gage, 6 feet long, shall be provided. A 50-ampere, uninsulated, electrical bronze, plier type clamp shall be soldered to each end of the wire. A ground terminal shall be bonded to the frame.

3.19 Identification, instruction and shipping plates.

3.19.1 Identification marking. Each pump unit shall be identified in accordance with MIL-STD-130 on a plate conforming to MIL-P-514, type as applicable. The plate shall be attached to the pump unit by screws, rivets, or bolts in a location where the plate will be visible and legible but not interfere with the operation and maintenance of the pump unit.

3.19.2 Identification of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump. The pump inlet and outlet shall have black painted arrows denoting the direction of flow.

3.19.3 Instruction plates. Each pump unit shall be equipped with instruction plates, including diagrams, warnings, and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the pump unit. Plates shall be conform to MIL-P-514, type optional. Two plates with the warning sign, "NO SMOKING WITHIN 50 FEET (15.2M)" that is visible from a distance greater than 50 feet shall be attached, one on each side, of the pump unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

3.19.4 Performance curve plate. A performance curve plate, 8 x 10-1/2 ±1/16-inch, shall be attached to the pump unit by screws, rivets, or bolts in a location where the plate will be visible during operation and start-up. The chart shall depict the performance curve for the speed required to meet the condition point as well as the condition point itself. Also, the legend shall list psi for liquids of 0.72 specific gravity, 0.85 specific gravity, and 1.0 specific gravity. The plate material shall be as specified in 3.19.3.

3.20 Lubrication. The pump unit shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Lubricating oil: MIL-L-2104 or MIL-L-46167
Grease: MIL-G-10924 or MIL-G-23827

The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.20.1 Lubrication fittings. The lubrication fittings shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.20.2 Pressure-release device. A pressure release device shall be provided where use of pressure lubricating equipment will damage grease seals of other parts.

3.21 Cleaning, treatment, and painting. All exterior surfaces normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. Unless otherwise specified (see 6.2), top coat shall be color No. 30277. Engine manifolds, exhaust pipes, mufflers and other parts subject to operating temperatures above 350 °F (177 °C) shall be painted with paint conforming to MIL-P-14105, color No. 33448.

3.22 Storage box. A storage box, constructed of steel not less than 15 gage thickness, shall be rigidly mounted on the front of the base. The storage box shall have an overlapping, lockable lid to prevent the entry of dirt and rain. The lid shall be secured at one edge by a continuous hinge conforming to MS35829, size as applicable, and shall be held in the closed position by not less than two fasteners. The box shall be of the largest possible size without interfering with operation and maintenance of the pump unit (minimum size 15 x 15 x 25 inches). The bottom of the box shall be provided with drain holes.

3.23 Battery box assembly. The battery box shall be weatherproof and made of material resistant to battery electrolyte and shall be designed to rigidly restrain two 12-volt batteries conforming to MIL-B-83769. The battery box shall be provided with a lid securely fastened to the box and designed to be easily removed for access to and observation of the battery cells and terminals. Battery cables shall be color coded, red for positive and black for negative. A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

pictorial battery connection diagram shall be provided. The diagram shall be readable while the batteries are being serviced. The box shall also provide gas venting of the battery cells and contain two holes for drainage. Electrolyte shall be provided.

3.24 Electromagnetic interference. Unless otherwise specified (see 6.2), the unit shall be equipped for the suppression of electromagnetic interference in accordance with MIL-STD-461 for engine-driven-equipment.

3.25 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable, with 1 percent copper 8-quinolinolate (by weight), based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26 Dimensions. The overall dimensions of the pump unit shall not exceed the following:

Length:	300 inches
Width:	80 inches
Height (assembled):	150 inches
Height (prepared for shipment):	80 inches

3.27 Weight. The weight of the pump unit shall not exceed 16,000 pounds, uncrated; 20,000 pounds, crated.

3.28 Workmanship.

3.28.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing process shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.28.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided where necessary and all bolts, nuts, and screws shall be tight and shall have full thread engagement. Prior to assembly, threads on all screwed fittings shall be coated with sealing compound conforming to MIL-S-45180. Bolts shall protrude not more than two full threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

3.28.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weldments, including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. All welding is to be produced by ASME code qualified welders. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Welds shall be in accordance with ASME Boiler and Pressure Vessel Code.

3.28.4 Welders. Before assigning any welder or welding operator to welding work, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed for the type of welding to be performed and that such qualification is effective as defined by the ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications. In the event of evidence of poor welds, the Government reserves the right to require testing of any welder or welding operator. All test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.28.5 Machine work. Tolerance and gages for metal fit shall conform to commercial industry standards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and tests of 4.6. Presence of one or more defects or failure of any tests shall be cause for rejection of the first article unit.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sample size shall be determined by using MIL-STD-105, tables I and IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects in 4.5. Presence of one or more defects shall be cause for rejection of the lot.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.6.1 and 4.6.2.2. Failure of any test shall be cause for rejection of the lot.

4.5 Examination. Each pump unit shall be examined for the defects given below:

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration in the applicable operating and storage environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the pump unit (see 3.3.4).
106. Design not as specified (see 3.4).
107. Interchangeability not as specified (see 3.5).
108. Construction not as specified (see 3.11).
109. Construction materials not as specified (3.11.1).
110. Seals and bearings not as specified (see 3.11.2).
111. Impellers not as specified (see 3.11.3).
112. Suction and discharge connections not as specified (see 3.11.4).
113. Casing vent lines not as specified (see 3.11.5).
114. Tapered pins not as specified (see 3.11.6).
115. Engine or accessories not as specified. Each incorrect accessory shall constitute one defect (see 3.12).
116. Engine enclosure not as specified (see 3.12.1).
117. Connect/disconnect clutch not as specified (see 3.12.2).
118. Automatic engine/pump control not as specified (see 3.13).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

119. Operator controls not as specified (see 3.13.1).
120. Manual engine speed control not as specified (see 3.13.2).
121. Automatic discharge pressure control not as specified (see 3.13.3).
122. Pressure ranges not as specified (see 3.13.4).
123. Instrument panel not as specified (see 3.14).
124. Fuel system not as specified (see 3.15).
125. Skid mount not as specified (see 3.16).
126. Transportability requirements not met (see 3.17 thru 3.17.5).
127. Grounding assembly not as specified (see 3.18).
128. Identification marking not as specified (see 3.19.1).
129. Pump inlet and outlet identification not as specified (see 3.19.2).
130. Instruction plates not as specified (see 3.19.3).
131. Performance curve plate not as specified (see 3.19.4).
132. Lubrication not as specified (see 3.20).
133. Lubrication fittings not as specified (see 3.20.1).
134. Lubrication fittings pressure-release device not as specified (see 3.20.2).
135. Finish treatment and painting missing, incomplete, or not as specified (see 3.21).
136. Storage box not as specified (see 3.22).
137. Battery box not as specified (see 3.23).
138. Fungus and moisture resistance treatment not as specified (see 3.25).
139. Dimensions specified are exceeded (see 3.26).
140. Weight specified is exceeded (see 3.27).
141. Fabrication not as specified (see 3.28.1).
142. Bolted connections not as specified (see 3.28.2).
143. Welding not as specified (see 3.28.3).
144. Welders certification not available (see 3.28.4).
145. Machine work not as specified (see 3.28.5).

4.6 Tests.

4.6.1 Hydrostatic test. Each pressure casing shall be hydrostatically tested in accordance with API Std 610.

4.6.2 Performance.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and liquid fuel at 60 °F having a specific gravity of 0.85. Water at a temperature of between 60 and 80 °F may be used as the test fluid. All performance and operational tests specified herein which require engine to be in operation shall be performed using referee grade diesel fuel conforming to MIL-F-46162 as the engine fuel.

4.6.2.2 Performance test. The pump unit shall be operated as specified in API Std 610, performance test. The performance test shall also include a net positive suction head required test in accordance with API Std 610.

4.6.2.3 Operational test and functional test. The pump unit shall be operated for 4 hours at the rated conditions specified in 3.12.1. The pump unit shall be operated as required during this period

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

to verify the functional operation of the automatic controls. The pump shall be examined during the operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component or inability of the pump to deliver the minimum capacities specified herein shall constitute failure of this test.

4.6.2.4 Tilted position. The pump unit shall be operated for not less than 30 minutes while it is positioned 5 degrees from level along the longitudinal centerline of the engine. Evidence of malfunction or misalignment of components shall constitute failure of this test.

4.6.2.5 Environmental tests. The pump unit shall be tested in accordance with MIL-STD-810 for the following:

a. Method 501.3, high temperature, procedure II. Maximum nonoperating temperature is 155 °F. Maximum operating temperature is 135 °F.
b. Method 502.3, low temperature, procedure I. Minimum nonoperating temperature is -60 °F. Minimum operating temperature is -25 °F.
c. Method 500.3, low pressure (altitude), procedure II. Maximum operating temperature of 100 °F at 6000 feet altitude.

A waiver of this test will be permitted when the contractor supplies certification that the pump unit is capable of operating in the specified environmental conditions. If certification of performance is provided and accepted, the contractor will provide a warranty to correct any malfunctions caused by operating in the conditions specified in 3.6.

4.6.2.6 Safety. The pump unit shall be evaluated for safety requirements throughout testing as specified in 4.6.2. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.7 Human factors engineering inspection. The human factors engineering characteristics shall be inspected during testing. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.8 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474, 5.1 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load; MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with 3.9 and MIL-STD-1474 provisions shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The pump unit shall be tested to determine conformance with 3.24. Nonconformance to 3.24 shall constitute failure of this test.

4.6.2.10 Transportability tests.

4.6.2.10.1 Tiedown provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.17.1. Inability to meet the requirements of 3.17.1 shall constitute failure of this demonstration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

4.6.2.10.2 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.17.2. Inability to meet the requirements of 3.17.2 shall constitute failure of this demonstration.

4.6.2.10.3 Air transport test. The pump unit shall be tested in accordance with MIL-STD-1791 to prove conformance to 3.17.3. Inability to meet the requirements of 3.17.3 shall constitute failure of this test.

4.6.2.10.4 Rail impact test. The first article only shall be tested in accordance with MIL-STD-810 to prove conformance to 3.17.4. Inability to meet the requirements of 3.17.4 shall constitute failure of this test.

4.6.2.10.5 Highway transport test. The pump unit shall be tested to prove conformance to 3.17.5. Inability to meet the requirements of 3.17.5 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article packaging inspection.

4.7.1.1 Examination. The first article packaging shall be examined for the defects specified in MIL-P-10603 and for the additional defects specified in 4.7.2.2 herein.

4.7.1.2 Tests. The testing of first article packaging shall be in accordance with MIL-P-10603.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Examination. Each pack shall be examined for the applicable defects listed in MIL-P-10603. Additionally, each pack shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

146. Wood/plywood not treated as specified for level A (see 5.3).
147. Cooling system not preserved as specified for level A (see 5.4.1).
148. Suction and discharge connection openings not protected with plywood as specified for level A (see 5.4.1.1).
149. Void areas of skid mount not closed off with treated plywood as specified for level A (see 5.4.1.2).
150. Crate cover not as specified for level A (see 5.5.1.1).
151. Crate cover not attached to skid mount as specified for level A (see 5.5.1.1).
152. Consolidation containers, when required, not as specified for level A (see 5.5.1.2).
153. Consolidation container not secured to the skid mount as specified for level A (see 5.5.1.2).
154. Packing not as specified by the referenced document for level C (see 5.5.2).
155. Marking incorrect, missing or illegible for levels A (see 5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The first article pack may be accomplished utilizing either the first article pump unit or a production pump unit. If the first article pump unit is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article pump unit and production pump units.

5.2 General. Special considerations shall be afforded this product because of an anticipated extended period of storage under unfavorable environmental conditions. The preservation measures, the application of preservatives, the materials and methods required for packing, and the detailed marking of the containers shall be administered in a manner to assure strict compliance to the requirements specified or referenced herein.

5.3 Wood preservation. All lumber and plywood used in the packaging of this product shall be treated with the water-repellent preservative copper naphthenate or copper 8-quinolinolate in accordance with TT-W-572, composition C or D.

5.3.1 Surface protection. All metal surfaces that come in direct contact with the treated wood/plywood shall be protected with noncorrosive barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall completely cover the contact surfaces and shall be secured in place, when required, with tape conforming to PPP-T-60.

5.3.2 Fasteners. All fasteners used in the fabrication of any component of a wooden container, all fasteners used in the blocking, bracing, and anchoring of the item to any components of a wooden container together shall be hot-dip galvanized fasteners.

5.4 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.4.1 Level A. Each complete pump unit shall be preserved in accordance with the level A requirements of MIL-P-10603 except that type P-14 preservative shall be used in lieu of type P-10, and the cooling system shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153, and that consolidation containers, when required, shall be in accordance with PPP-B-601, overseas type.

5.4.1.1 Suction and discharge connection openings. In addition to the barrier material application required in MIL-P-10603 to seal certain openings, each suction and discharge connection opening shall be protected with a sheet of 3/8-inch thick plywood (see 5.3). The plywood sheet shall extend, around the perimeter, a minimum of 1 inch beyond the flange and shall be attached to the connection with a minimum of 4 equally spaced metal mechanical fasteners.

5.4.1.2 Skid mount. Void areas between the supporting cross members of the skid mount and, as applicable, void areas at the ends of the skid, shall be closed off with 3/4-inch thick

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

plywood shields (see 5.3). The plywood shall be secured to the cross members with metal mechanical fasteners or clamps of such type, size, and quantity to assure positive positioning of the shield. Alternatively, strategically positioned wood cleats may be attached to the plywood shields to perform the same function. When portions of a component of the pump unit extend below the plywood shield, provisions shall be made in the shields to accommodate those portions and they shall be protected by a wood cleated, 3/4-inch thick plywood enclosure that is secured to the shield. Nominal 2-inch by 4-inch wood joists shall be attached to the plywood shields, as required, to add rigidity and to act as a batten when butting (splicing) two pieces of plywood together.

5.4.1.3 Electrolyte. Each 3.5 liter acid resistant electrolyte container (8 containers per pump unit) shall be placed in a snug-fitting fiberboard box conforming to PPP-B-636, weather-resistant.

5.4.2 Level C. Each complete pumping unit shall be preserved as specified for level A except the boxes shall be domestic type or class, whichever is applicable.

5.5 Packing. Packing shall be level A or C, as specified (see 6.2).

5.5.1 Level A.

5.5.1.1 Cover. Each pump unit, preserved as specified in 5.4, shall be packed in a close-fitting, rectangular, wood cleated, plywood cover assembly that encloses the sides, ends, and top of the unit. The cover shall be constructed in accordance with MIL-C-104, type II, class 2, style a, except that the wood skid base shall be omitted and that the lower frame members of the sides and ends shall be positioned flush with the bottom edge of the plywood sheathing. The crate shall be constructed to allow removal of the complete assembly using standard slings.

5.5.1.2 Consolidation containers. When consolidation containers are required, they shall be secured to the skid mount, under the cover, with flat steel strapping conforming to QQ-S-781, finish B.

5.5.1.3 Cover attachment. The cover assembly specified in 5.5.1.1 shall be attached to the skid mount by one of the following methods:

a. By the attachment of wood cleats to the top of the skid mount. Such cleats shall be positioned around the perimeter of the skid and the cover shall be bolted to the cleats. Space permitting, cleats shall be the same size as lower frame members of cover, sides and ends.
b. By the attachment of metal angles or channels to the top of the skid mount. Such angles or channels shall be positioned around the perimeter for the skid and the cover shall be bolted to the angles or channels.
c. By the attachment of series of clips on the outside of the cover at the bottom of the sides and ends. Such clips shall be bolted to the skid mount.
d. By the combination of any or all of the above.
e. The spacing of the fasteners attaching the cover by any of the above methods shall be in accordance with MIL-C-104 for the type II crate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

f. Fasteners required of a and b, above, to attach material to the top of the skid mount shall not interfere with the fasteners attaching the cover.

5.5.1.4 Electrolyte. The electrolyte, preserved as specified in 5.4, shall be packed (2 fiberboard boxes) in a snug-fitting wooden container conforming to PPP-B-601, overseas type. Blocking and bracing shall conform to MIL-STD-1186 and strapping shall be in accordance with ASTM D 3953 and D 4675, type 1 or 2, grade l or 2.

5.5.2 Level C. Each pump unit, preserved as specified in 5.4, shall be packed in accordance with MIL-STD-1190.

5.6 Marking. In addition to any special or other markings required of the contract or order (see 6.2), each crate shall be marked in accordance with MIL-STD-129, including all appendices as applicable for overseas shipment for use by a Military Command. Interior and exterior packing lists are required in accordance with MIL-STD-129, appendix F. The packing list on the outside of the crate shall be inserted in a moistureproof envelope and stapled to the side of the crate. The envelope shall then be covered with a fiberboard cover stapled to the crate. The crate surfaces on which markings are to be applied shall be prepared in accordance with MIL-STD-129, appendix A. All markings shall be l-inch stencils, overcoated (weatherproofed) with a clear coating compound. When the pump units are to be included as part of a set, assembly or module, special care must be exercised to ensure each container is marked as specified. Bar code markings shall be utilized.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump unit will be used primarily to pump fuel through a pipeline in cold (-25 °F) and hot (135 °F) temperatures. The operating conditions will range from dry to humid with frequent periods of blowing sand.

6.2 Acquisitions requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for the first article inspection and approval (see 3.2, 4.3 and 6.3).
d. When data for air transportability is required (see 3.17.3).
e. Color of finish coat if other than as specified (see 3.21).
f. When electromagnetic interference suppression is not required (see 3.24).
g. Time frame required for submission of first article pack (see 5.1 and 6.5).
h. Level of preservation and packing required (see 5.4 and 5.5).
i. Special marking required (see 5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

j. Additional required data to be furnished to the manufacturer (see appendix).
k. When a transportability report is required (see 6.6 and 6.6.1).

6.3 First article. When a first article inspection is required, the item(s) should be a first produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the pump unit in accordance with this specification.

6.6 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P. 70-38 for the pump unit(s), 95 days prior to delivery of the first article pump unit(s).

6.6.1 Rotary wing aircraft evaluation. The contractor should be in contact with Natick Research, Development, and Engineering Center to get an evaluation of the pump unit for rotary wind air transportability. The evaluation is required prior to completion of the transportability report.

6.7 Superseding data. This specification supersedes PD 4320-0015, Pumping Assembly, Diesel-Engine-Driven, Centrifugal, 800 GPM at 1800 Feet Total Head, dated 6 January 1985.

6.8 Figures. Figure 1a and 1b showing the general layout of the unit are for illustration only. Location of pump unit components are optional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

6.9 Subject term (key word) listing.

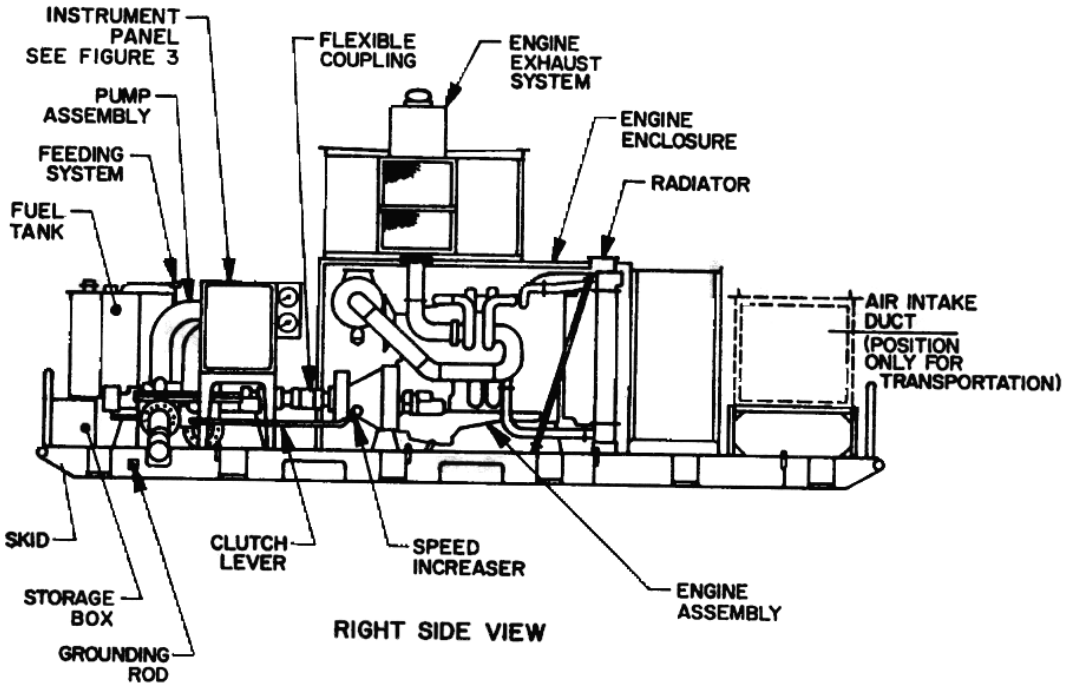
Controls
Engine
Engine/pump controller
Indicator
Instrument panel
Mode control
Pump
Skid-mounted

Custodian:
Army – ME

Preparing activity:
Army – ME

Project 4320-A227

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.



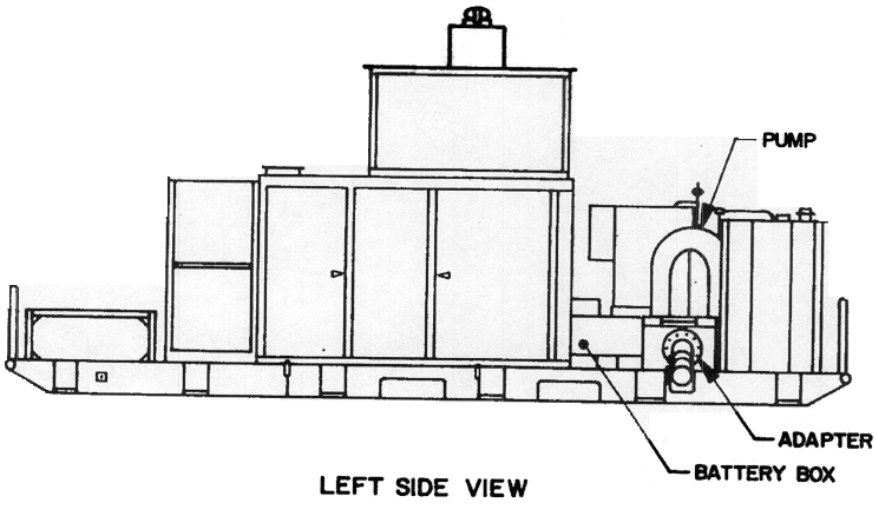


FIGURE 1a. Pump unit – major components.

X-4967

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

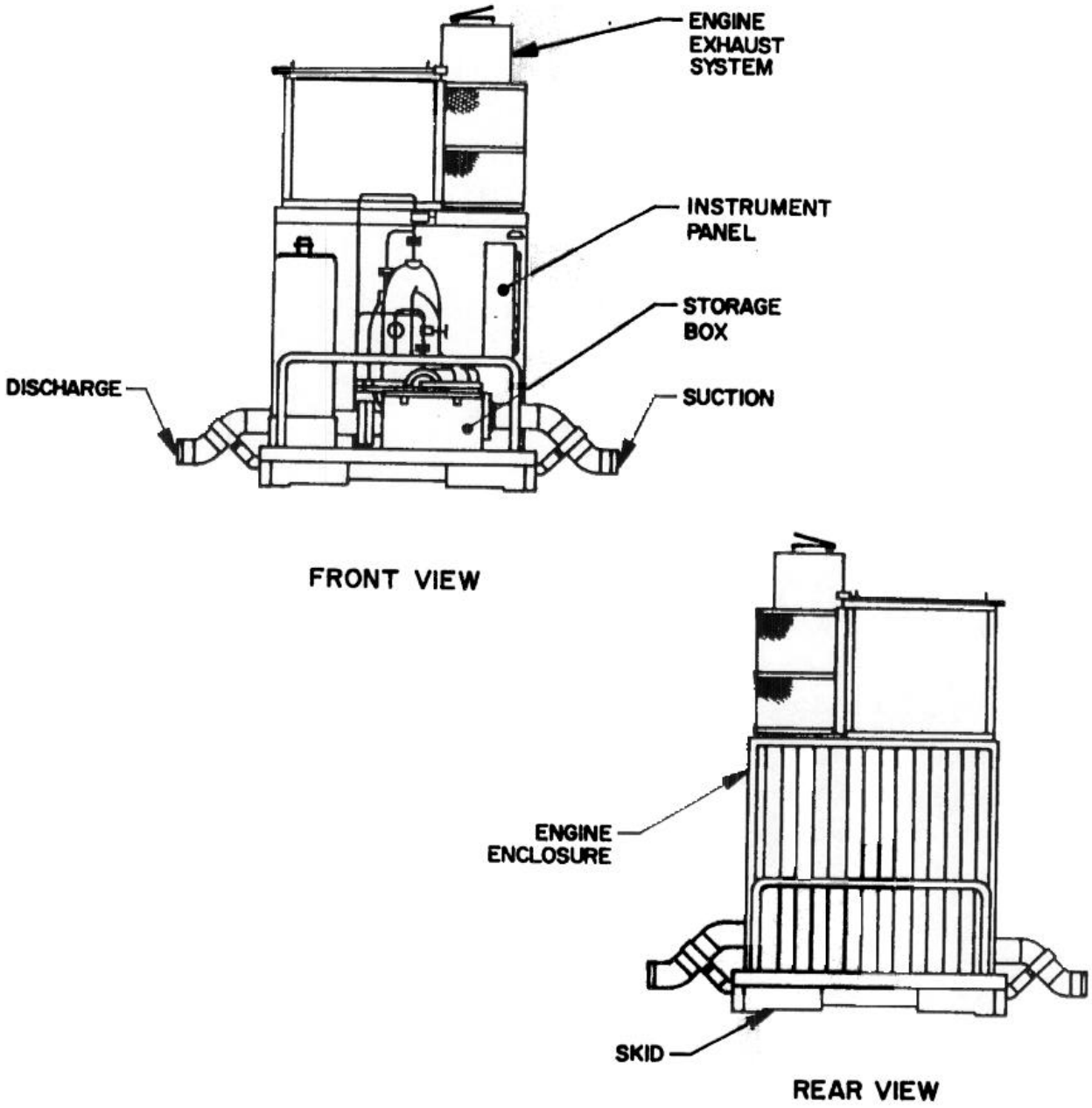
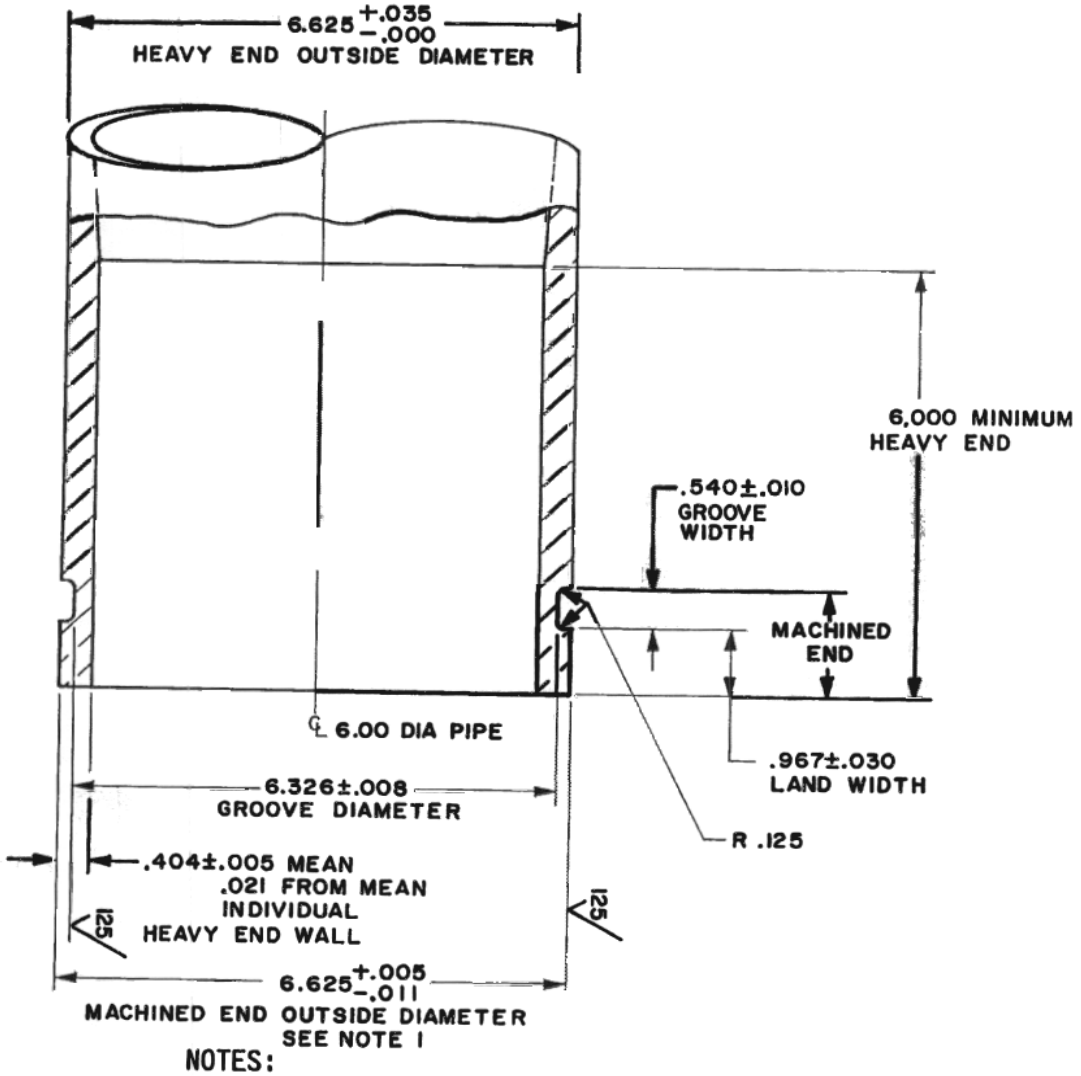


FIGURE 1b. Pump unit - major components - Continued.

X-4968

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.



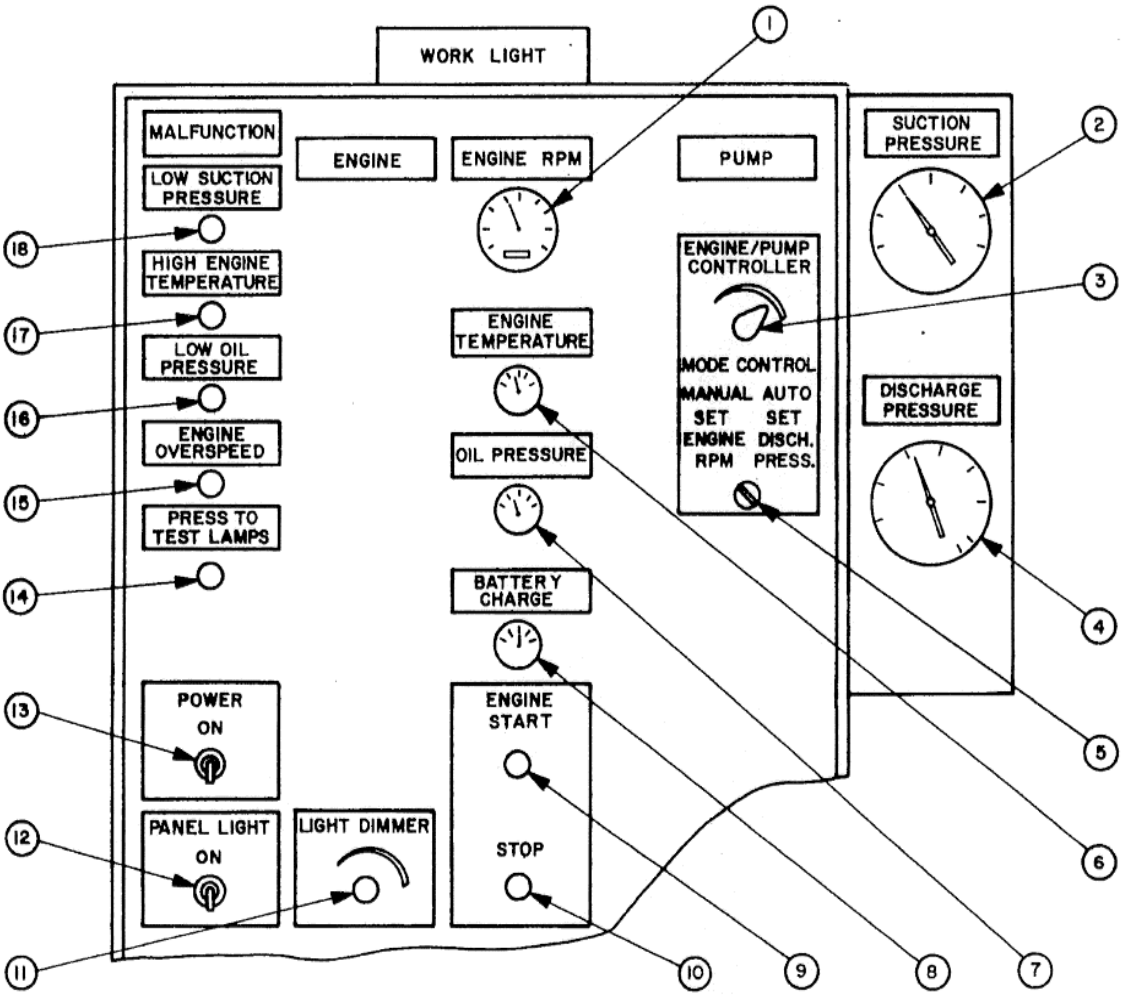
NOTES:

1. MACHINED DIMENSIONS ARE BASED ON INTERNAL DIAMETER CHUCKING OF PIPE.
2. ALL DIMENSIONS ARE IN INCHES.

FIGURE 2. Dimensions for 6 inch IPDS single groove end.

X-4834A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.



LEGEND

1. ENGINE TACHOMETER
2. SUCTION PRESSURE GAUGE
3. ENGINE/PUMP CONTROL POTENTIOMETER
4. DISCHARGE PRESSURE GAUGE
5. ENGINE MODE CONTROL SWITCH
6. ENGINE COOLANT TEMPERATURE
7. ENGINE OIL PRESSURE GAUGE
8. BATTERY VOLTMETER
9. START SWITCH
10. PULL TO STOP SWITCH
11. LIGHT DIMMER SWITCH
12. PANEL LIGHTS SWITCH
13. POWER SWITCH
14. TEST LAMPS SWITCH
15. ENGINE OVERSPEED ALARM
16. LOW OIL PRESSURE ALARM
17. HIGH ENGINE TEMP. ALARM
18. LOW SUCTION PRESSURE ALARM

FIGURE 3. Instrument panel.

X-4969

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.

APPENDIX

SPECIFIC REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix provides specific required data in design and construction of the pump unit to meet purchaser's data requirements.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. SUBMISSION

See API Std 610 Paragraph Number	Data requirements
2.1.11	Parallel operation is anticipated.
2.3.2.11	Pressure gauge connections are required.
2.7.1.12	Plan 31 shall be used as a guide for mechanical seal piping plan.
2.8.3.4	Torsional vibration analysis is required. Torsional critical shall not coincide with two, three, or four times running speeds.
3.2.2 and 3.2.7	Couplings shall be stainless steel disk type and dynamically balanced.
3.3.1.12	Pump pedestal shall be sufficiently rigid to mount without grouting.
3.5.2.11	Flush fluid (diesel fuel) is flammable.
4.3.1.2	Purchaser review of test procedures is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:07Z
Check the source to verify that this is the current version before use.